Aditya Yadav

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SKILLS

Languages: Python, Java, C#, JavaScript, TypeScript, Go, C/C++, SQL, **Frontend:** React, Next.js, HTML, CSS, **Backend & APIs:** FastAPI, Spring Boot, Node.js, Express.js, REST APIs, GraphQL, OpenAPI, **AI/ML Frameworks:** TensorFlow, PyTorch, Scikit-learn, XGBoost, Keras, **NLP & LLM Tools:** Hugging Face, LangChain, OpenAI APIs, FAISS, Vector Databases, RAG Pipelines, **Data Engineering & Analytics:** Pandas, NumPy, Power BI, Matplotlib, Airflow, **Databases:** MySQL, PostgreSQL, MongoDB, Redis, ClickHouse, **Cloud, DevOps & MLOps:** AWS (ECS Fargate, ECR, Lambda, API Gateway, S3, CloudWatch), Docker, Kubernetes, CI/CD, GitHub Actions, MLflow, Model Monitoring

EXPERIENCE

Atlassian | **AI/ML Engineer** | USA | Jun 2025 – Present

- Engineered **AI-driven workflow automation** powering platforms for **10K+ enterprise users**, while processing **500K+ weekly ticketing records** through high-throughput **Python** pipelines.
- Optimized **NLP pipelines** using **Hugging Face**, **LangChain**, and **FAISS** index optimizations, boosting semantic search relevance by **25%** and accelerating vector retrieval by **35%**.
- Architected highly available **FastAPI microservices** handling **15K+ daily requests**, conducting load testing and latency profiling to ensure strict SLA compliance.
- Deployed and monitored **5+ production AI microservices** on **AWS** (**ECS, ECR, Docker, S3**) with **CI/CD** pipelines, increasing overall system reliability and throughput by **20%**.
- Automated reporting and monitoring workflows, cutting manual effort by **30%** while collaborating cross-functionally to translate ambiguous AI feature requests into testable engineering scope.

Zoho | **Software Engineer** | India | Aug 2022 – Aug 2024

- Built scalable **Python** automation and **ETL** pipelines processing **1M+ records** for CRM applications serving **100K+ active users**, replacing manual data-sync scripts with scheduled **Airflow DAGs**.
- Integrated production **ML models** for task/lead prediction, improving operational workflow efficiency by **32%**; owned the full lifecycle from feature engineering through model deployment and monitoring.
- Expanded **FastAPI backend services** and **PostgreSQL schemas** for new AI analytics modules, designing versioned schema migrations that shipped with zero downtime across a multi-tenant production environment.
- Reduced API response times by **27%** via query optimization (indexing, query plan analysis) and **Redis caching** for high-traffic endpoints, directly improving dashboard load times for end users.
- Cut manual validation overhead by **40%** by building automated data-quality checks and reporting scripts, catching schema drift and malformed records before they reached production.
- Partnered with product and QA cross-functionally to ship AI features on aggressive release cycles, writing **integration tests** that reduced regression bugs in the CRM's analytics module.
- Mentored **2 junior engineers** on **Python** best practices, code review standards, and testing discipline, contributing to a measurable drop in post-merge defects on the team

EDUCATION

Stevens Institute of Technology | Master of Science | Computer Science | May 2026

- AWS Certified ML Engineer – Associate
- TensorFlow Developer Professional Certificate

PROJECTS

Catalyst Map | Next.js, TypeScript, Groq LLM API, Tailwind CSS | GitHub

- Architected a real-time market catalyst research platform that ingests event data and uses an **LLM pipeline** to surface and summarize market-moving signals for users.
- Wrote end-to-end **Vitest test coverage** suites specifically targeting adversarial cases—malformed JSON, provider timeouts, and empty responses—to lock in reliability before shipping.

MorphArc | React, TypeScript, MediaPipe, Three.js, Web Workers | GitHub

- Built a real-time hand-tracking interface that maps gesture input to 3D object manipulation, running **MediaPipe inference** off the main thread via **Web Workers** to keep the UI responsive.
- Validated tracking accuracy and rotation stability with **66 automated unit/integration tests**, covering edge cases like partial occlusion and rapid gesture transitions.

RunRoute | Next.js, TypeScript, Tailwind CSS, Mapbox GL JS | GitHub

- Designed a geospatial route-planning app that generates running routes deterministically from user-set constraints (target distance, elevation preference, avoid-highway filters).
- Built the route generation algorithm around **graph-based pathfinding** over map segment data, rather than naive straight-line snapping, to produce realistic runnable loops.